Venkata Lakshmi Parimala Pasupuleti

AI Engineer · LangGraph · Multi-Agent · RAG · LLM Evaluation · Observability · AWS · Python · TypeScript · Production AI · CI/CD Eval Gates · McLean VA

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PROFESSIONAL SUMMARY

AI engineer with 8 years shipping production systems and 3+ years hands-on agentic AI — based in McLean, VA. Built **AgentProbe**: a 3-agent LangGraph runtime (Orchestrator, RAG Tracer, Auto Fix) with a **5-metric LLM-as-judge eval engine**, CI/CD regression gates blocking failing runs before production, and a React observability dashboard tracking per-run metrics, latency, and retrieval drift — the production mindset the JD calls for, live on GitHub. Built **Leasel**: production RAG over 1,240 legal PDFs (93.6% precision, 87.9% F1) with a pass@k eval harness and regression gate — not a prototype. At Entrata: **ran AWS S3/RDS/Glue at production scale** (100% PHI compliance, zero incidents), shipped 20+ full-stack modules across 1M+ records, led Docker containerization cutting execution time 50%. M.S. Data Analytics Engineering, GMU (GPA 3.87, May 2026).

WORK EXPERIENCE

George Mason University
Aug 2024 – Aug 2025

Research Associate — ML Pipelines · Anomaly Detection · Statistical Modeling

- Built end-to-end ML pipelines (Python, R, 190+ countries) achieving **$R^2 = 0.977$**; applied Isolation Forest anomaly detection across 3 research pipelines surfacing anomalies manual review missed; automated statistical reporting with Python + Quarto driving **40% improvement** in stakeholder comprehension.

Entrata
Jan 2021 – Aug 2024

Senior Software Engineer — Production AI · AWS · Quality Gates · Observability · Full-Stack

- Shipped Entrata's **first production AI pipeline across 1.6M real financial records** (Python, PyTorch, OpenAI API) with output validation, quality gates, and HITL review before customer exposure — zero production incidents at launch.
- Achieved **100% PHI compliance** across all client data by migrating to AWS S3/RDS/Glue with encrypted pipelines — zero security incidents; established the observability and testing standards that caught issues before production across 20+ modules.
- Led **Docker microservices containerization** (50% execution time reduction); **defined technical standards** for 20+ production modules (Python, PostgreSQL, RabbitMQ) across 1M+ records and 100K+ MAU — 99.5% uptime; mentored 5+ engineers, 300+ code reviews.

Entrata
Jan 2019 – Dec 2020

Software Engineer — API Integrations · Compliance Automation · Data Pipelines

- Built compliance automation processing **millions of records** with deterministic rules engines and full audit trails; built API integrations and webhooks across property management systems; tuned **200+ SQL queries** achieving 30% query performance improvement.

Entrata
Jan 2017 – Dec 2018

Associate Engineer — Full-Stack · Python · PostgreSQL · REST APIs

- Built and shipped full-stack features and SQL data layers (Python, JavaScript, PostgreSQL) for production workflows; 0→1 prototyping through production rollout across 100K+ unit portfolios.

PROJECTS

AgentProbe: Multi-Agent AI Eval & Observability Framework
MULTI-AGENT · LLM-AS-JUDGE · PRODUCTION EVAL

Python · LangGraph · FastAPI · ChromaDB · OpenAI · Anthropic · Docker · GitHub Actions — github.com/venkatalakshmiparimala/AgentProbe

- 3-agent LangGraph runtime (Orchestrator, RAG Tracer, Auto Fix) with tool calling and inter-agent state handoffs; **5-metric LLM-as-judge eval engine** (factual accuracy, hallucination, retrieval, completeness, compliance); **CI/CD regression gates blocking failing runs before production** on every commit; 15-endpoint FastAPI backend; React **observability dashboard** with per-run metrics, latency tracking, and retrieval drift — production-quality automation, not a prototype.

Leasel: Agentic RAG Compliance Pipeline
AGENTIC RAG · PASS@K EVAL

Python · LangGraph · ChromaDB · Deterministic Rules Engine · LLM-as-Judge · Pass@k Eval — github.com/venkatalakshmiparimala/Lease_complaince_rag

- **93.6% precision, 87.9% F1** on 1,240 complex legal PDFs — production RAG pipeline: extraction, chunking, embedding, retrieval, deterministic rules engine for fully auditable outputs; pass@k eval harness (pass@1 81.7%, pass@3 88.3%) with LLM-as-judge scoring and CI/CD regression gate. Production-quality system with measurable eval, not a demo.

AI on FHIR: HIPAA-Compliant Medical Records NLP Platform
HEALTHCARE

Python · Flask · React · FHIR APIs · BERT · LLMs · Embeddings · Docker — github.com/venkatalakshmiparimala/AI_on_FHIR

- ~60% reduction in clinical retrieval time — NLP platform translating natural-language queries (EN/ES) into FHIR API calls via LLMs + BERT; HIPAA-compliant RBAC with end-to-end logging and access controls across structured medical records.

EDUCATION

M.S. Data Analytics Engineering — George Mason University — May 2026 · GPA 3.87

ML systems, statistical modeling, data engineering, AI pipelines, system design, anomaly detection

B.Tech Computer Science & Engineering — JNTU Anantapur — 2013 – 2017

TECHNICAL SKILLS

Languages & Backend: Python, TypeScript, JavaScript, SQL, R; FastAPI, Flask, Django, REST APIs, webhooks, RabbitMQ, event-driven distributed systems

AI & Agentic Systems: LangGraph, LangChain, Multi-Agent Orchestration, Agentic AI, RAG Pipelines, Tool Calling, LLM-as-Judge, LLM Evaluation, AI observability, HITL Feedback Loops, Vector Stores (ChromaDB, pgvector), OpenAI API, Anthropic API, PyTorch, BERT, NLP, prompt engineering, MCP servers (familiar), AWS Bedrock (familiar, extending AWS experience)

Infrastructure & Cloud: AWS (S3, EC2, RDS, Glue), Docker, Kubernetes, CI/CD (GitHub Actions), Microservices, PostgreSQL, MySQL, HIPAA/PHI compliance, scalable architecture, Agile/Scrum, Git, Linux